🕐 Clocks & Calendars — Deep Dive (SSC CGL Reasoning)

> Detailed companion to the Clocks & Calendars row in SSC-CGL-ASO-Formulas-Tricks-Master.md. Syllabus: Reasoning (Para 13.10.1 / 13.11.2). Read each idea, cover the box, redo the worked example.

Contents

1. Clock basics — the geometry
2. Angle between the hands
3. When do the hands coincide / oppose / make 90°?
4. Fast / slow / faulty clocks
5. Calendars — odd days
6. Leap years
7. Finding the day of any date
8. Repeating calendars
9. Practice set + answers

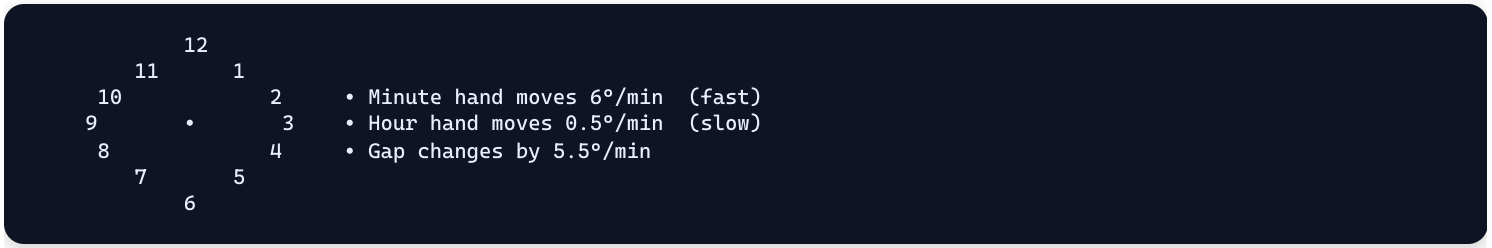
PART A — CLOCKS

1. Clock basics — the geometry

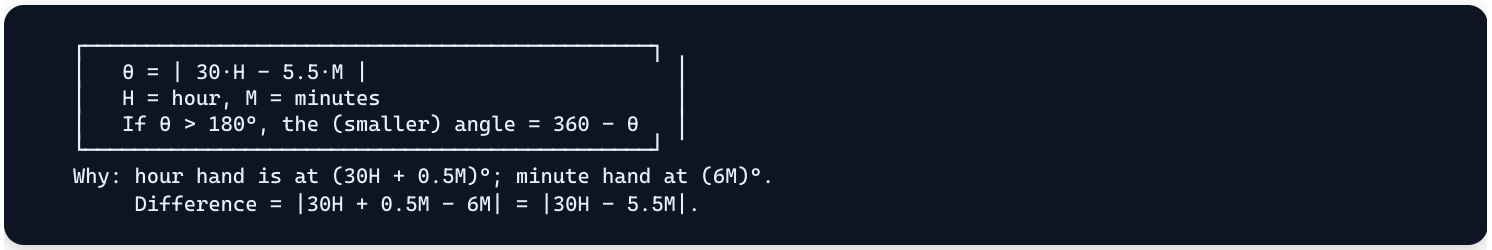
```
A clock face = 360°, divided into 12 hours and 60 minutes.
┌──────────────────────────────────────────┐
│ Full circle ............... 360°        │
│ 1 hour space (12 → 1) ..... 360/12 = 30° │
│ 1 minute space ............ 360/60 = 6° │
│ Each hour = 5 minute-spaces (30° = 5 × 6°)│
└──────────────────────────────────────────┘
```

Hand	Speed	Per minute
Minute hand	360° in 60 min	**6° per minute**
Hour hand	360° in 12 h = 720 min	**0.5° per minute**
Relative gain	minute over hour	**6 − 0.5 = 5.5° per minute** ⭐

🔑 **The single most useful number:** the minute hand gains **5.5° every minute** on the hour hand. Almost every clock question reduces to "*how long to gain X degrees?*" → time = X ÷ 5.5 min.



2. Angle between the hands ⭐



Worked example 1 — angle at 3:40

- θ = |30×3 − 5.5×40| = |90 − 220| = |−130| = **130°**.

Worked example 2 — angle at 4:20

- θ = |30×4 − 5.5×20| = |120 − 110| = **10°**.

Worked example 3 — reflex check at 9:15

- θ = |30×9 − 5.5×15| = |270 − 82.5| = 187.5° → greater than 180°
- smaller angle = 360 − 187.5 = **172.5°**.

 Speed tip: at H o'clock exactly (M=0), angle = 30H (take the smaller side). E.g. 5:00 → 150°.

3. When do the hands coincide, oppose, or make 90°?

The hands start a "race"; the minute hand catches up at **5.5°/min**.

Event	Angle to cover	Times in 12 hours	Gap between events
Coincide (0° apart)	0° / 360°	**11 times**	every 720/11 = **65 5/11 min**
Opposite (180° apart)	180°	**11 times**	every 65 5/11 min
Right angle (90°)	90° (twice/cycle)	**22 times**	—

```
WHY 11 (not 12) overlaps in 12 hours:
In 12 h the minute hand laps the hour hand only 11 times
(the 12 o'clock overlap is shared by start & end).
→ overlaps at 12:00, 1:05 5/11, 2:10 10/11, … about every 65.45 min.
```

Worked example 4 — after 4 o'clock, when do hands coincide?

- At 4:00 the hour hand leads by 4×30 = 120°.
- Minute hand must gain 120° at 5.5°/min → time = 120 ÷ 5.5 = **21 9/11 min past 4** (≈ 4:21:49).

Worked example 5 — between 4 and 5, when are hands at right angles (90°)?

- Need gap = 90°. Start gap (at 4:00) = 120°.
- **First 90°:** reduce 120 → 90, gain (120−90)=30° → 30÷5.5 = 5 5/11 min → **4:05 5/11**.
- **Second 90°:** go past coincidence to 90° on other side: gain (120+90)=210° → 210÷5.5 = 38 2/11 → **4:38 2/11**.

4. Fast, slow & faulty clocks

Situation	How to handle	□ Trick
Clock **gains** x min/day	runs fast; shows later time	add error over elapsed days
Clock **loses** x min/day	runs slow; shows earlier time	subtract error
Hands coincide every X min	if X ≠ 65 5/11, clock is **faulty**	compare to 65.4545…
Total gain/loss	rate × (number of days/hours)	proportion

Worked example 6 — fast clock

> A clock gains 5 minutes every hour. If set correct at 8:00 a.m., what does it show at the real 8:00 p.m. (same day)?

- Real time elapsed = 12 hours → gain = 5 × 12 = 60 min = 1 hour.
- Clock shows **9:00 p.m.**

Worked example 7 — "faulty" detection

> The hands of a clock coincide every 64 minutes of correct time. Is it fast or slow?

- Correct clocks coincide every 65 5⁄11 ≈ 65.45 min. This one does it in 64 min (sooner) → it is running **fast**.
- Gain per coincidence = 65 5⁄11 − 64 = 16⁄11 min, so over a day it can be scaled up.

PART B — CALENDARS

5. Calendars — the "odd days" method ⭐

```
┌──────────────────────────────────────────┐
│ ODD DAYS = days left over after counting │
│ complete weeks (remainder when ÷ 7).     │
│ The day of the week shifts by the odd days.│
└──────────────────────────────────────────┘
1 ordinary year = 365 days = 52 weeks + 1 → 1 odd day
1 leap year     = 366 days = 52 weeks + 2 → 2 odd days
```

Period	Odd days	□ Trick
Ordinary year	**1**	"next year same date = next weekday"
Leap year	**2**	jumps two weekdays
100 years	**5**	(76 ordinary + 24 leap → 76+48 = 124 ; 124 mod 7 = 5)
200 years	**3**	(5×2 = 10 → 3)
300 years	**1**	(5×3 = 15 → 1)
400 years	**0**	(and every multiple of 400) ⭐

```
ODD-DAY COUNT PER MONTH (memorise the remainders, value mod 7)

┌────────┬──┬────────┬──┬────────┬──┐
│ Jan 31 │ 3│ May 31 │ 3│ Sep 30 │ 2│
│ Feb 28 │ 0│ Jun 30 │ 2│ Oct 31 │ 3│   (Feb 29 in a leap yr → 1)
│ Mar 31 │ 3│ Jul 31 │ 3│ Nov 30 │ 2│
│ Apr 30 │ 2│ Aug 31 │ 3│ Dec 31 │ 3│
└────────┴──┴────────┴──┴────────┴──┘

□ 30-day months give 2 odd days; 31-day months give 3; Feb gives 0 (or 1 if leap).
```

6. Leap years

```
┌──────────────────────────────────────────┐
│ Leap year rule:                          │
│ • divisible by 4              → leap     │
│ • BUT century years (÷100)    → NOT leap … │
│ • … UNLESS also ÷ 400         → leap     │
└──────────────────────────────────────────┘

✅ Leap:  2008, 2012, 2024, 2000, 2400
❌ Not :  2009, 1900, 2100, 2200, 2300
```

 Trick: 1900 is **not** a leap year (÷100 but not ÷400); 2000 **is** (÷400). This single fact is a very common trap.

7. Finding the day of any date ⭐

Method (odd-days from a reference): add odd days for completed centuries + years + months + date, take mod 7, map to a weekday.

```
WEEKDAY CODE (from total odd days, mod 7)
0 → Sunday   1 → Monday   2 → Tuesday   3 → Wednesday
4 → Thursday 5 → Friday   6 → Saturday
```

Worked example 8 — what day was 15 August 1947? Step 1: 1600 yrs → 0 odd days. (use nearest 400-multiple before the year) Step 2: 300 yrs (1601–1900) → 1 odd day. Step 3: 46 complete years (1901–1946): leap years among them = 46÷4 = 11 leaps, 35 ordinary. odd days = 35×1 + 11×2 = 35 + 22 = 57 → 57 mod 7 = **1**. Step 4: year 1947, months Jan–Jul + 15 days of Aug: Jan31 Feb28 Mar31 Apr30 May31 Jun30 Jul31 = 212 days; + 15 = 227 days → 227 mod 7 = **3**. Step 5: total odd days = 0 + 1 + 1 + 3 = 5 → **Friday**. ✅ **15 August 1947 was a Friday.**

```
COUNT-UP DIAGRAM
[centuries: 0+1] + [years 1901-46: 1] + [this year's months+date: 3]
         = 5  →  Friday
```

Worked example 9 — same date, next year

> If 1 Jan 2025 is Wednesday, what is 1 Jan 2026?

- 2025 is an ordinary year → 1 odd day → shift by 1 → **Thursday**.

> And 1 Jan 2027? (2026 ordinary → +1) → **Friday**.

8. Which year has the same calendar?

Case	Calendar repeats after	□ Trick
Ordinary year after ordinary	+6 or +11 years (commonly **+11**)	sum of odd days = 7
Leap year	repeats after **28 years**	full cycle
Quick rule	add years until cumulative odd days = 0 (mod 7)	—

Worked example 10

> Which year will have the same calendar as 2024 (a leap year)?

- Leap-year calendars repeat every 28 years → **2052**.

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9. Practice set

> Cover the answers. Aim: < 40 seconds each.

#	Question	Answer				
1	Angle between hands at **2:30**?	\	30·2 − 5.5·30\	= \	60−165\	= **105°**
2	Angle at **6:00**?	180° (straight line)				
3	How many times do the hands coincide in **a day (24 h)**?	**22 times** (11 per 12 h)				
4	Hands coincide every 65 5⁄11 min — true gap in a **correct** clock?	yes, ≈ **65.45 min**				
5	A clock loses 10 min/day. After 6 days it shows?	**1 hour slow** (10×6=60 min)				
6	Odd days in a **leap year**?	**2**				
7	Is **2100** a leap year?	**No** (÷100, not ÷400)				
8	If 1 Jan 2024 is Monday, 1 Jan 2025 is?	2024 leap → +2 → **Wednesday**				
9	Odd days in **400 years**?	**0**				
10	Day on **26 January 1950** (Republic Day)?	**Thursday**				
11	Between 3 and 4 o'clock, hands coincide at?	3:00 gap 90° → 90÷5.5 = **3:16 4⁄11**				
12	A leap year's calendar repeats after?	**28 years**				

```
Q11 working:  at 3:00 hour hand leads by 90°.
Minute hand gains 90° at 5.5°/min → 90 ÷ 5.5 = 16 4⁄11 min past 3.
```

✅ **Two numbers carry this whole topic: 5.5°/min** (clocks) and **odd days mod 7** (calendars). Master those and every question becomes a one-line calculation. Back to the main formula sheet ».